

Doctor Consultation Workflow

Backend Changes Documentation

Version: 1.0 | Date: June 2026 | Scope: HM-System backend only

1. Purpose

This document describes backend changes for the Doctor Consultation workflow. The goal is to let doctors open a consultation without mutating queue status, save clinical data in one atomic step, and keep appointments and patient_queue consistent across doctor, reception, nurse, OPD, and pharmacy modules.

2. Problem Solved (Before vs After)

Before

- Opening consult required queue/add and queue/start (status mutations).
- Saving required queue_id and a 3-step chain: add, start, complete.
- PUT /queue/complete accepted clinical JSON but did not persist it.
- PUT /appointments/{id}/status could not go scheduled to completed.

After

- GET /consultations/appointment/{id} loads context read-only.
- POST /consultations/save and PUT /queue/complete complete in one request.
- PATCH /consultations/appointment/{id} completes with queue optional.
- Clinical fields persist on appointments (symptoms, diagnosis, notes, follow_up_date).
- PUT /appointments/{appointment_id}/status supports scheduled/waiting to completed with queue sync.

3. New API Endpoints

Doctor Consultations (/consultations)

Tag: Doctor Consultations | Router: Routers/doctor_consultation_router.py

- GET /consultations/appointment/{appointment_id} - Read-only context (appointments:view)
- POST /consultations/save - Atomic save with queue ensured (appointments:update)
- PATCH /consultations/appointment/{appointment_id} - Queue-optional save (appointments:update)

Doctor Patient Queue - Extended (/queue)

Tag: Doctor Patient Queue | Router: Routers/doctor_patient_queue_router.py

- PUT /queue/complete - NEW: Atomic save by appointment_id (same logic as POST /consultations/save)
- PUT /queue/complete/{queue_id} - ENHANCED: Persists clinical body; returns appointment in response

Doctor Appointments - Updated (/appointments)

- PUT /appointments/{appointment_id}/status - scheduled/waiting -> completed allowed; syncs queue if row exists

4. API Request / Response Examples

POST /consultations/save | PUT /queue/complete

Request:

```
{
  "appointment_id": 17,
  "clinical": {
    "symptoms": "Fever, cough",
    "diagnosis": "Viral URI",
    "notes": "Rest and fluids",
    "follow_up_date": "2026-07-15"
  }
}
```

Response 200:

```
{
  "success": true,
  "message": "Consultation saved",
  "appointment": { "id": 17, "status": "completed", ... },
  "queue": { "id": 42, "status": "completed", "appointment_id": 17 }
}
```

GET /consultations/appointment/{appointment_id}

Response 200:

```
{
  "success": true,
  "appointment": { ... patient + visit fields ... },
  "queue": { "id": 42, "status": "waiting", ... } | null,
  "prescriptions": [ ... recent for patient ... ],
  "lab_orders": [ ... for this appointment ... ]
}
```

PATCH /consultations/appointment/{appointment_id} (queue optional)

Request body (clinical only):

```
{
  "symptoms": "Fever",
  "diagnosis": "Viral URI",
  "notes": "Rest",
  "follow_up_date": "2026-07-15"
}
```

Behavior:

- Does NOT create patient_queue row.
- If queue row exists today -> marks queue completed + saves clinical.
- If no queue row -> only appointment completed + clinical saved.

5. Which Endpoint to Use

Use POST /consultations/save or PUT /queue/complete when the doctor should finish the visit in one step and reception/nurse must see patient_queue completed (queue row created internally if missing).

Use PATCH /consultations/appointment/{id} when queue is optional for the doctor (no new queue row; only updates existing queue if checked in).

Use PUT /queue/complete/{queue_id} when you already have queue_id from reception check-in.

Use PUT /appointments/{id}/status with status=completed for status-only completion (no clinical body); syncs queue if row exists.

6. Internal Service Flow

Atomic save (POST /consultations/save, PUT /queue/complete)

- save_consultation_service() in Services/doctor_consultation_service.py
- 1. Load and lock appointment; reject completed/cancelled.
- 2. ensure_queue_for_appointment() - get or create today's queue row.
- 3. finalize_consultation() - clinical fields + queue completed + appointment completed.
- 4. Single database commit.

Queue-optional save (PATCH /consultations/appointment/{id})

- complete_appointment_consultation_service() in Services/doctor_appointment_service.py
- complete_queue_for_appointment_if_exists() - only if row exists today.
- Otherwise appointment-only completion with clinical fields.

Queue helpers (Services/doctor_patient_queue_service.py)

- find_queue_for_appointment_today()
- ensure_queue_for_appointment()
- complete_queue_for_appointment_if_exists()
- finalize_consultation()
- _apply_clinical_to_appointment()

7. Files Added

- Routers/doctor_consultation_router.py
- Services/doctor_consultation_service.py
- Schemas/doctor_consultation_schema.py
- alembic/versions/a9b0c1d2e3f4_appointment_symptoms.py

8. Files Modified

- main.py - registered consultation_router
- Models/opd_billing.py - added appointments.symptoms column
- Services/doctor_helpers.py - appointment_to_dict includes symptoms
- Services/doctor_appointment_service.py - VALID_TRANSITIONS, PATCH service logic, status queue sync
- Services/doctor_patient_queue_service.py - clinical persist, helpers, complete_consultation fix
- Routers/doctor_patient_queue_router.py - PUT /queue/complete, enhanced complete response
- Schemas/doctor_appointment_schema.py - AppointmentConsultationUpdate

9. Database Changes

- Table: appointments
- New column: symptoms (TEXT, nullable)
- Existing columns used: diagnosis, notes, follow_up_date
- No new tables. No new status enum values.
- Migration: alembic upgrade head (revision a9b0c1d2e3f4)

10. Appointment Status Transitions (VALID_TRANSITIONS)

```
scheduled -> waiting | completed | cancelled
waiting    -> in_progress | completed | cancelled
in_progress -> completed | cancelled
completed  -> (none)
cancelled   -> (none)
```

11. Cross-Module Consistency

- Doctor dashboard reads appointments.status
- Reception and nurse read patient_queue
- Pharmacy POST /prescriptions requires appointment.status == completed
- Atomic save updates both appointment and queue in one transaction
- PATCH queue-optional path may leave no queue row (nurse/reception queue empty for that visit)

12. Legacy Queue APIs (Unchanged)

- GET /queue/today
- PUT /queue/start/{queue_id}
- GET /queue/current
- POST /queue/request-next
- POST /queue/add - deprecated; use POST /receptionist/check-in/{appointment_id}

13. Error Codes (Common)

- 404 - Appointment or queue not found / not doctor's patient
- 400 - Already completed, cancelled, or invalid status transition
- 409 - Patient already has another queue entry with same doctor today (atomic save)

14. Deployment Checklist

- 1. Run: alembic upgrade head
- 2. Restart FastAPI server
- 3. Verify Swagger tags: Doctor Consultations, Doctor Patient Queue
- 4. Test POST /consultations/save from scheduled appointment
- 5. Test PATCH /consultations/appointment/{id} without prior check-in
- 6. Test PUT /queue/complete with same body as POST /consultations/save